

ECO101 – Introduction to Microeconomics

Lecture 7: Perfect Competition

Medium-length study notes based on the uploaded lecture slides.

Topics covered

1. Perfect competition and its characteristics
2. Price takers
3. Economic profit and revenue: TR, TC, MR, AR
4. Demand, price and revenue under perfect competition
5. Firm's output decision
6. Profit maximization using marginal analysis
7. Temporary shutdown decision and shutdown point
8. Firm's supply curve
9. Short-run market supply and short-run equilibrium
10. Short-run outcomes: break-even, profit and loss
11. Entry and exit
12. Long-run equilibrium
13. Bottled-water exercise

1. Perfect Competition

Perfect competition is a market structure where many firms sell identical products to many buyers. There are no restrictions on entry, established firms have no advantage over new firms, and buyers and sellers are well informed about prices.

The important idea is that an individual firm is too small relative to the whole market to control the market price.

- Many firms + identical products
- Free entry and exit
- No established-firm advantage
- Buyers and sellers are well informed

2. Price Takers

A perfectly competitive firm is a **price taker**. It accepts the market price rather than choosing its own price. Since the product is homogeneous, charging more than the market price causes consumers to buy from other firms.

The individual firm's production is insignificant compared with total market production, so changing its own output does not meaningfully change the market price.

- Firm takes P as given
- Higher-than-market price $\rightarrow$ customers buy elsewhere
- The firm cannot influence market price

3. Economic Profit and Revenue

The firm's goal is to maximize economic profit. Total revenue is the money received from selling output, while total cost is the cost of producing it.

Marginal revenue is the additional total revenue from selling one more unit. Average revenue is revenue per unit

sold.

- Profit = TR – TC
- TR = P × Q
- MR = $\Delta TR / \Delta Q$
- AR = TR / Q

4. Demand, Price and Revenue in Perfect Competition

Market demand and market supply determine the market price. Once the market price is determined, an individual competitive firm can sell any quantity it chooses at that price. Therefore, the firm's demand curve is horizontal at the market price and is perfectly elastic.

Because each additional unit is sold at the same market price, marginal revenue equals price.

- Firm demand is horizontal
- Firm demand is perfectly elastic
- MR = P
- In perfect competition: P = MR = AR

5. The Firm's Output Decision

The competitive firm chooses the quantity of output that maximizes economic profit. One method is to compare total revenue and total cost at different output levels. The lecture example shows the firm maximizing profit at 9 sweaters per day; at 4 and 12 sweaters it breaks even, while outputs below 4 or above 12 produce an economic loss.

- Choose Q that gives the highest economic profit
- Break-even: TR = TC
- Profit > 0 when TR > TC
- Profit < 0 when TR < TC

6. Profit Maximization: Marginal Analysis

A faster method is to compare marginal revenue with marginal cost. If MR is greater than MC, producing one more unit adds more revenue than cost, so profit rises. If MR is less than MC, another unit reduces profit. Profit is maximized where MR = MC, provided the firm is operating in the relevant range.

For a competitive firm, MR = P, so the key rule becomes P = MC at the profit-maximizing output.

- MR > MC → increase output
- MR < MC → decrease output
- MR = MC → profit-maximizing output
- Perfect competition: P = MR = MC at Q*

7. Temporary Shutdown Decision

A firm can make an economic loss in the short run because fixed costs must still be paid. If the loss is expected to be temporary, the firm compares the loss from producing with the loss from shutting down.

If the firm shuts down, Q = 0 and it avoids variable costs, but it still pays total fixed cost (TFC). If producing generates enough revenue to cover variable cost, continuing to operate can minimize the loss.

- Shutdown loss = TFC
- Continue producing when revenue covers variable cost
- If $P < AVC$, shutdown
- If $P \geq AVC$, produce the profit/loss-minimizing output

8. The Shutdown Point

The shutdown point occurs at the minimum point of the average variable cost curve. Below minimum AVC, the firm's revenue cannot cover its variable cost, so producing would create a loss greater than its fixed cost. At the shutdown point, the firm is indifferent between shutting down and operating at the relevant minimum-AVC output.

- Shutdown if $P < \text{minimum AVC}$
- At shutdown: loss = TFC
- If $\text{minimum AVC} < P < ATC \rightarrow$ produce but incur a loss
- If $P \geq ATC$ at the chosen output $\rightarrow$ economic profit is non-negative

9. Firm's Supply Curve

The firm's short-run supply curve shows how its profit-maximizing quantity changes as the market price changes. It is derived from the firm's marginal cost curve above the minimum AVC point.

When price rises, the firm moves upward along its MC curve and supplies more. When price falls below minimum AVC, it shuts down and supplies zero.

- Short-run firm supply = MC above minimum AVC
- $P < \text{minimum AVC} \rightarrow Q = 0$
- Higher $P \rightarrow$ higher quantity supplied

10. Market Supply in the Short Run

The short-run market supply curve adds the quantities supplied by all firms at each price. The number of firms is fixed in the short run.

For example, if there are 1,000 identical firms, market quantity supplied at a given price equals 1,000 times the quantity supplied by one firm.

- Market Q_s = sum of all firms' Q_s
- With 1,000 identical firms: $Q_{\text{market}} = 1,000 \times Q_{\text{firm}}$
- Number of firms stays fixed in the short run

11. Short-Run Equilibrium

Market demand and short-run market supply determine the equilibrium price and market output. Each individual firm takes that price as given and chooses its profit-maximizing quantity using $P = MR = MC$.

In the lecture example, at $P = \$20$ each firm produces 8 sweaters. With 1,000 identical firms, market output is 8,000 sweaters per day.

- Market equilibrium: demand intersects short-run supply
- Firm takes equilibrium P as given
- $Q_{\text{market}} = \text{number of firms} \times Q_{\text{firm}}$

12. Short-Run Profits, Break-Even and Losses

In the short run, a firm can earn economic profit, break even, or incur an economic loss.

Profit can be written using average total cost. If price is greater than ATC, profit is positive. If price equals ATC, economic profit is zero. If price is below ATC, the firm has an economic loss.

- Profit = $(P - ATC) \times Q$
- $P > ATC \rightarrow$ economic profit
- $P = ATC \rightarrow$ zero economic profit / break-even
- $P < ATC \rightarrow$ economic loss

13. Entry and Exit

Economic profit encourages new firms to enter a competitive market. Entry increases market supply, which pushes the market price down and reduces existing firms' profit.

Economic loss encourages firms to leave. Exit decreases market supply, which pushes price up and reduces the losses of remaining firms.

Importantly, temporary profit or loss does not automatically cause entry or exit; the lecture emphasizes persistent expected profit or loss.

- Profit $\rightarrow$ entry $\rightarrow$ supply increases $\rightarrow$ price falls $\rightarrow$ profit falls
- Loss $\rightarrow$ exit $\rightarrow$ supply decreases $\rightarrow$ price rises $\rightarrow$ loss falls
- Entry and exit stop when economic profit is zero

14. Long-Run Equilibrium

A competitive market reaches long-run equilibrium when economic profit and economic loss have been eliminated and entry and exit have stopped. The key mechanism is adjustment in the number of firms.

Entry increases market output but decreases each firm's output in the lecture example. Exit decreases market output but increases each remaining firm's output.

- Long-run equilibrium: zero economic profit
- No further entry or exit
- Persistent profit attracts entry
- Persistent loss causes exit

15. Important Formula Sheet

Profit: $\pi = TR - TC$

Total revenue: $TR = P \times Q$

Average revenue: $AR = TR / Q$

Marginal revenue: $MR = \Delta TR / \Delta Q$

Profit using ATC: $\pi = (P - ATC) \times Q$

Economic loss: $Loss = TFC + TVC - TR$

Shutdown condition: $P < \text{minimum AVC} \rightarrow$ shut down

Operating with loss: $\text{minimum AVC} \leq P < ATC$

Perfect competition: $P = MR = AR$

Profit-maximizing condition: $MR = MC$; therefore, in perfect competition, $P = MC$

16. Lecture Exercise: Bottled Water

The lecture gives:

$$TC = 40 + 4Q + Q^2$$

$$MC = 4 + 2Q$$

Market price $P = \text{Tk.}15$

There are 200 firms.

1. Profit-maximizing output

For a competitive firm, set $P = MC$:

$$15 = 4 + 2Q \rightarrow 11 = 2Q \rightarrow Q = 5.5 \text{ units.}$$

2. Short-run profit

$$\text{At } Q = 5.5: TC = 40 + 4(5.5) + (5.5)^2 = 92.25.$$

$$TR = 15(5.5) = 82.50.$$

$$\text{Profit} = 82.50 - 92.25 = -\text{Tk.}9.75.$$

3. Shut down or continue?

$TVC = 4Q + Q^2 = 52.25$, so $AVC = 52.25/5.5 \approx \text{Tk.}9.50$. Since $P = \text{Tk.}15 > AVC$, the firm should continue operating in the short run even though it makes a loss.

4. Long run

If losses persist, firms will exit the market. Market supply falls, price rises, and the remaining firms' losses decrease. Entry/exit continues until zero economic profit is restored in long-run equilibrium.

Quick Exam Logic

Question	Rule
What does a competitive firm do with price?	Takes market price as given
What is the firm's demand curve?	Horizontal / perfectly elastic
What is MR?	$MR = P$
How to find Q^* ?	Set $MR = MC$; in perfect competition, $P = MC$
When does the firm shut down?	When $P < \text{minimum } AVC$
When does it break even?	$P = ATC$ at Q^*
When does it earn profit?	$P > ATC$ at Q^*
What causes entry?	Persistent economic profit
What causes exit?	Persistent economic loss
Long-run competitive equilibrium?	Zero economic profit; entry and exit stop